

THE NEURO-SYMBOLIC CONCEPT LEARNER: INTERPRETING SCENES, WORDS, AND SENTENCES FROM NATURAL SUPERVISION

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ABSTRACT

We propose the Neuro-Symbolic Concept Learner (NS-CL), a model that learns visual concepts, words, and semantic parsing of sentences without explicit supervision on any of them; instead, our model learns by simply looking at images and reading paired questions and answers. Our model builds an object-based scene representation and translates sentences into executable, symbolic programs. To bridge the learning of two modules, we use a neuro-symbolic reasoning module that executes these programs on the latent scene representation. Analogical to human concept learning, the perception module learns visual concepts based on the language description of the object being referred to. Meanwhile, the learned visual concepts facilitate learning new words and parsing new sentences. We use curriculum learning to guide the searching over the large compositional space of images and language. Extensive experiments demonstrate the accuracy and efficiency of our model on learning visual concepts, word representations, and semantic parsing of sentences. Further, our method allows easy generalization to new object attributes, compositions, language concepts, scenes and questions, and even new program domains. It also empowers applications including visual question answering and bidirectional image-text retrieval.

1 INTRODUCTION

Humans are capable of learning visual concepts by jointly understanding vision and language (Fazly et al., 2010; Chrupała et al., 2015; Gauthier et al., 2018). Consider the example shown in Figure 1-I. Imagine that someone with no prior knowledge of colors is presented with the images of the red and green cubes, paired with the questions and answers. They can easily identify the difference in objects’ visual appearance (in this case, color), and align it to the corresponding words in the questions and answers (Red and Green). Other object attributes (e.g., shape) can be learned in a similar fashion. Starting from there, humans are able to inductively learn the correspondence between visual concepts and word semantics (e.g., spatial relations and referential expressions, Figure 1-II), and unravel compositional logic from complex questions assisted by the learned visual concepts (Figure 1-III, also see Abend et al. (2017)).

Motivated by this, we propose the neuro-symbolic concept learner (NS-CL), which jointly learns visual perception, words, and semantic language parsing from images and question-answer pairs. NS-CL has three modules: a neural-based perception module that extracts object-level representations from the scene, a visually-grounded semantic parser for translating questions into executable programs, and a symbolic program executor that reads out the perceptual representation of objects, classifies their attributes/relations, and executes the program to obtain an answer.

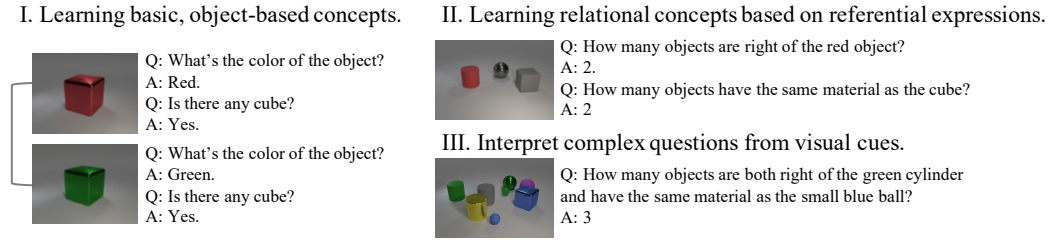


Figure 1: Humans learn visual concepts, words, and semantic parsing jointly and incrementally. **I.** Learning visual concepts (red vs. green) starts from looking at simple scenes, reading simple questions, and reasoning over contrastive examples (Fazly et al., 2010). **II.** Afterwards, we can interpret referential expressions based on the learned object-based concepts, and learn relational concepts (e.g., on the right of, the same material as). **III.** Finally, we can interpret complex questions from visual cues by exploiting the compositional structure.

NS-CL learns from natural supervision (i.e., images and QA pairs), requiring no annotations on images or semantic programs for sentences. Instead, analogous to human concept learning, it learns via curriculum learning. NS-CL starts by learning representations/concepts of individual objects from short questions (e.g., What's the color of the cylinder?) on simple scenes (≤ 3 objects). By doing so, it learns object-based concepts such as colors and shapes. NS-CL then learns relational concepts by leveraging these object-based concepts to interpret object referrals (e.g., Is there a box right of a cylinder?). The model iteratively adapts to more complex scenes and highly compositional questions.

NS-CL's modularized design enables interpretable, robust, and accurate visual reasoning: it achieves state-of-the-art performance on the CLEVR dataset (Johnson et al., 2017a). More importantly, it naturally learns disentangled visual and language concepts, enabling combinatorial generalization w.r.t. both visual scenes and semantic programs. In particular, we demonstrate four forms of generalization. First, NS-CL generalizes to scenes with more objects and longer semantic programs than those in the training set. Second, it generalizes to new visual attribute compositions, as demonstrated on the CLEVR-CoGenT (Johnson et al., 2017a) dataset. Third, it enables fast adaptation to novel visual concepts, such as learning a new color. Finally, the learned visual concepts transfer to new tasks, such as image-caption retrieval, without any extra fine-tuning.

2 RELATED WORK

Our model is related to research on joint learning of vision and natural language. In particular, there are many papers that learn visual concepts from descriptive languages, such as image-captioning or visually-grounded question-answer pairs (Kiros et al., 2014; Shi et al., 2018; Mao et al., 2016; Vendrov et al., 2016; Ganju et al., 2017), dense language descriptions for scenes (Johnson et al., 2016), video-captioning (Donahue et al., 2015) and video-text alignment (Zhu et al., 2015).

Visual question answering (VQA) stands out as it requires understanding both visual content and language. The state-of-the-art approaches usually use neural attentions (Malinowski & Fritz, 2014; Chen et al., 2015; Yang et al., 2016; Xu & Saenko, 2016). Beyond question answering, Johnson et al. (2017a) proposed the CLEVR (VQA) dataset to diagnose reasoning models. CLEVR contains synthetic visual scenes and questions generated from latent programs. Table 1 compares our model with state-of-the-art visual reasoning models (Andreas et al., 2016; Suarez et al., 2018; Santoro et al., 2017) along four directions: visual features, semantics, inference, and the requirement of extra labels.

For visual representations, Johnson et al. (2017b) encoded visual scenes into a convolutional feature map for program operators. Mascharka et al. (2018); Hudson & Manning (2018) used attention as intermediate representations for transparent program execution. Recently, Yi et al. (2018) explored an interpretable, object-based visual representation for visual reasoning. It performs well, but requires fully-annotated scenes during training. Our model also adopts an object-based visual representation, but the representation is learned only based on natural supervision (questions and answers).

Anderson et al. (2018) also proposed to represent the image as a collection of convolutional object features and gained substantial improvements on VQA. Their model encodes questions with neural

Models	Visual Features	Semantics	Extra Labels		Inference
			# Prog.	Attr.	
FiLM (Perez et al., 2018)	Convolutional	Implicit	0	No	Feature Manipulation
IEP (Johnson et al., 2017b)	Convolutional	Explicit	700K	No	Feature Manipulation
MAC (Hudson & Manning, 2018)	Attentional	Implicit	0	No	Feature Manipulation
Stack-NMN (Hu et al., 2018)	Attentional	Implicit	0	No	Attention Manipulation
TbD (Mascharka et al., 2018)	Attentional	Explicit	700K	No	Attention Manipulation
NS-VQA (Yi et al., 2018)	Object-Based	Explicit	0.2K	Yes	Symbolic Execution
NS-CL	Object-Based	Explicit	0	No	Symbolic Execution

Table 1: Comparison with other frameworks on the CLEVR VQA dataset, w.r.t. visual features, implicit or explicit semantics and supervisions.

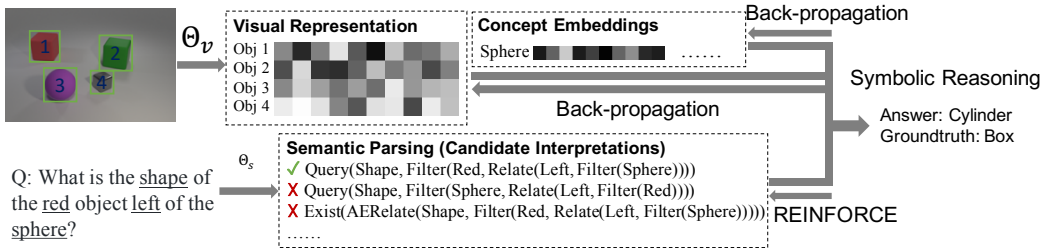


Figure 2: We propose to use neural symbolic reasoning as a bridge to jointly learn visual concepts, words, and semantic parsing of sentences.

networks and answers the questions by question-conditioned attention over the object features. In contrast, NS-CL parses question inputs into programs and executes them on object features to get the answer. This makes the reasoning process interpretable and supports combinatorial generalization over quantities (e.g., counting objects). Our model also learns general visual concepts and their association with symbolic representations of language. These learned concepts can then be explicitly interpreted and deployed in other vision-language applications such as image caption retrieval.

There are two types of approaches in semantic sentence parsing for visual reasoning: implicit programs as conditioned neural operations (e.g., conditioned convolution and dual attention) (Perez et al., 2018; Hudson & Manning, 2018) and explicit programs as sequences of symbolic tokens (Andreas et al., 2016; Johnson et al., 2017b; Mascharka et al., 2018). As a representative, Andreas et al. (2016) build modular and structured neural architectures based on programs for answering questions. Explicit programs gain better interpretability, but usually require extra supervision such as ground-truth program annotations for training. This restricts their application. We propose to use visual grounding as distant supervision to parse questions in natural languages into explicit programs, with *zero* program annotations. Given the semantic parsing of questions into programs, Yi et al. (2018) proposed a purely symbolic executor for the inference of the answer in the logic space. Compared with theirs, we propose a quasi-symbolic executor for VQA.

Our work is also related to learning interpretable and disentangled representations for visual scenes using neural networks. Kulkarni et al. (2015) proposed convolutional inverse graphics networks for learning and inferring pose of faces, while Yang et al. (2015) learned disentangled representation of pose of chairs from images. Wu et al. (2017) proposed the neural scene de-rendering framework as an inverse process of any rendering process. Siddharth et al. (2017); Higgins et al. (2018) learned disentangled representations using deep generative models. In contrast, we propose an alternative representation learning approach through joint reasoning with language.

3 NEURO-SYMBOLIC CONCEPT LEARNER

We present our neuro-symbolic concept learner, which uses a symbolic reasoning process to bridge the learning of visual concepts, words, and semantic parsing of sentences without explicit annotations